

Age Focused Analysis of Olympic Alpine Skiing

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I. INTRODUCTION

In recent years, the athletic competitive spirit has expanded far beyond the physical field of play. In an effort to remain competitive, and improve on-field performance, athletes, coaches, and managers have increasingly turned to analytical methods and tools in search of finding a differentiating edge. Although varying greatly in the degree and method of adoption, professional sports have increasingly expanded data analytics operations and capabilities. This increase in investment is not without purpose, gradually these teams and organizations have come to recognize the unique utility analytical methods offer in areas ranging from strategy development to scouting and signing decisions. These analytical frameworks have offered a way to better inform their decision making process which is crucial in a field driven by complex decision making in a noisy and uncertain environment. For these same reasons, the United States Olympic & Paralympic Committee (USOPC) has retained our services to expand their existing analytical tools and capabilities in an effort to improve Team USA's Olympic competitiveness.

II. USOPC BACKGROUND

The USOPC is a non-profit corporation with far more limited capital resources than other (non-Olympic) private for-profit athletic clubs. Operating within these constraints, the USOPC must decide when, how, and by what amount to fund promising athletes to maximize both the athlete's and Team USA's potential for Olympic success. These allocation decisions are extremely complex and involve the careful consideration of numerous underlying factors unique to the varying scales at which the USOPC funds athletes and sibling organizations. Further complicating the task is the sparsity in which athletes compete on the Olympic stage (every 4 years). This time lag makes it difficult to measure and gauge the success/progress of internal initiatives and overall competitiveness against other national committees. These two problems represent a prime target for analytical evaluation, and if modeled properly could yield significant value to the organization. Definitely solving these problems directly is beyond the scope of a single report. However, one way to begin the process is to start by simplifying the problem into more approachable sub-problems. It is with this general approach in mind that the USOPC tasked each of their practicum teams to analyze one specific Olympic discipline (our project will focus on Solo Alpine Skiing) with respect to the single factor of age. By better understanding the relationship between age and Olympic medal achievement, the USOPC will be able

to more effectively gauge their competitive potential, strength, and weaknesses in both past and future Olympic competitions. From a funding perspective, this improved understanding of the drivers and state of competitiveness within each discipline can inform future funding allocation decisions which are key to maximizing Team USA's medal output. Additionally, a through understanding of age at the individual level will give the USOPC insights on the most important windows for funding athletes in order to cultivate their peak athletic potential, and fielding athletes that are near the peak of their competitiveness for a given event.

III. ALPINE SKIING BACKGROUND

The discipline of Alpine skiing which is the primary focus of our analysis is comprised of five primary events Slalom, Giant Slalom, Super Giant Slalom (Super-G), and Downhill. Each of these five events challenge skiers to complete a run of a predetermined course in the fastest time possible while navigating through the specified checkpoints (gates). The differences between each event can be best explained by describing the two extremes, Slalom and Downhill with Giant Slalom and Super G falling in the middle of this range. The Slalom event is the shortest event, but requires more technical ability and precision due to the number of gates and turns present on the course. At the other end of the spectrum is Downhill which uses the largest course with minimal gates and turns instead focusing on overall speed and ability to handle multiple steep drops as the skier races towards the finish line. In addition to these five events, there is one additional competition included within the solo Olympic Alpine events list known the Super Combined/Combined. The Combined event although not part of international skiing competitions outside of the the Winter games, is composed of 2 runs of a Slalom event followed by a single run of Downhill the times of which are combined to determine a victor. Outside of Olympic competition, elite level alpine skiing events are hosted by the International Ski Federation (FIS) the governing body of competitive alpine ski racing. In order to understand a few of our modeling decisions there are a few other small details about alpine skiing to highlight that may be non-obvious to those unfamiliar with the sport. Firstly, All alpine skiing disciplines are physically demanding sports and are inherently prone to injury due to high racing speeds, course obstacles, and generally unforgiving course terrain and conditions. Although alpine skiing safety has advanced a long way in the modern era, it is still an important aspect of the sport to understand as it likely contributes to several of our analytical observations.

Secondly, alpine skiing course layouts are updated year to year so runs from the same competition, event, and resort in different years are likely not directly comparable due to small changes in course design. Thirdly, the FIS breaks alpine skiers into three classes based upon their competitive ability youth, juniors, and elite. Due to a lack of significant data volume for youth and junior athletes all of our analysis (outside of forecast Milan 2026 will focus on elite level skiers). Fourthly, Elite alpine skiers typically specialize in a single discipline, but the most elite alpine skiers have been successful and generally competitive in multiple events. Fifthly, the FIS as well as our presented web application tracks an event listed as "Overall". Although listed as an event, it is really a world composite ranking of all Elite level skiers that aggregates their ability in all five disciplines. Finally, in alpine skiing especially among top level skiers that are pushing the limits of a course and their ability often "ski-out" resulting in a "did not finish" (DNF) final position. This behavior is to be expected of athletes pushing the boundaries of what is possible in the sport. With this in mind, because no points are awarded for DNF finishes, there is a direct trade off in consistency and performance as one DNF run disqualifies you from the competition immediately even in events with multiple runs. This aspect of the sport becomes important as we attempt to evaluate skier performance and navigate the trade-off between valuating consistency or "gold-or-nothing" strategies. This quick overview should provide sufficient general context of alpine skiing to follow our subsequent analysis.

IV. PROJECT OVERVIEW

The goal of our project is to provide a comprehensive analysis on the impact of age with respect to Olympic medal achievement among the Winter Olympic discipline of solo alpine skiing. Throughout this process, we also aim to answer the following specific questions posed by the USOPC:

- 1) How does the health of Team USA's pipeline compare to other top medal winning countries through the primary lens of age?
- 2) Can a country's percentage of athletes competing at an Olympic games while in their peak age ranges in their respective sports predict medal success?
- 3) Which athletes who competed in Beijing 2022 will be hitting their peak age in their respective sports in Milan 2026?
- 4) Can the number of young, "pre-peak" athletes that competed in Beijing predict medal success for their countries in Milan 2026?
- 5) How many promising athletes who did not compete in Beijing, but will reach peak age in Milan does Team USA and other countries have in their pipeline?

Through this project report and the accompanying web application we aim to provide the USOPC definitive answers to each of these questions while also providing a thorough explanation and justification of the methods used to reach our conclusions.

V. METHODOLOGY

The primary objective of the project provided by the USOPC was to determine the relationship between age and medal achievement. However, in order to frame this in a more concrete and empirically verifiable way, we first need to test if there was any relationship between medal achievement and age so as not to make any fatal early assumptions. Then, if such a significant relationship is proven to exist, we would then need to quantify the extent and characteristics of the identified relation. Ideally, the age and medal achievement relationship would need to be further analyzed with respect to other potential explanatory variables to verify that the identified relationship was not instead caused by a collection of confounding or highly correlated variable(s). However, due to the limited nature of our data set as well as the particular characteristics of alpine skiing competition, there were limited other relations to explore beyond the athletes age a given competition and the corresponding competition results. In order to provide a complete and comprehensive analysis, each of these identified potential confounding or significant factors are discussed below and categorized into three groups according to their underlying characteristics. For the purposes of this methodology we will define these three groups of features according to the following three criterion:

- 1) Knowable (Known vs. Unknown): The feature and its variations can clearly be identified and known to exist. Known features encompass all identified features relevant to the analysis even those which are covered in research beyond this work.
- 2) Quantifiable (Measurable vs. Immeasurable): The feature which is known to exist can be quantified by some form of object, construct, datum, or measurement either directly or by a mathematically justified indirect measurement or approximation.
- 3) Obtainable (Available vs. Unavailable): The feature which is known to exist, and to be quantifiable, can be reasonably stored, measured, or otherwise captured in non-prohibitively cost effective manor. The definition lacks something in terms of specificity as to what is "cost effective", but we believe it best to leave this open ended as the definition likely depends on the entity conducting the analysis.

As suggested in the definitions above, a given feature's membership within one of these sets can imply membership in other sets. For example all "Obtainable" features assume both "Quantifiable" and "Knowable" membership. Conversely, Unknown features imply they are disjoint from each of the three sets. We will begin by discussing alternative features which are Knowable, Quantifiable, Obtainable, and as a result were evaluated further within our analysis.

A. Known, Measurable, and Available Features

In the case of any sport, there is often a tendency to compare the past greats to players in the modern era. However, these comparisons are often extremely difficult as sports are

constantly evolving in numerous dimensions from regulations to field of competitors. Among these dimensions, one that was found to be identifiable, reasonably measurable, and available within our given data set was the level of competition difficulty in alpine skiing over time. By quantifying and evaluating the progression of the competitive field and its difficulty historically we will be able to better understand if an athlete's particular place in time plays a significant role in determine medal achievement. The conclusions of this analysis aim to firstly provide some insight into another feature and perspective of modeling medal achievement. However, it additionally would provide context to the analysis of age. If the two variables were found to be significantly correlated, it would demonstrate that the relation between age and medal achievement isn't as direct or straight forward as we had hoped, and would suggest our results in general are unreliable or incomplete. Alternatively, if they do not share a strong pairwise relation, but are both significant in explaining medal achievement, they could jointly better model medal achievement, and potentially generalize far better than a standalone model. The driving push for this line of analysis was due to the perceived large differences in medal achievement age between post and pre 2000's event data during our exploratory data analysis. The significant variance between these two groups also drove us to seek out other potential explanatory factors or changes within the sport that could provide insight. Eventually we discovered that a key regulation change occurred after the 2002 season in an effort to improve athlete safety. Large changes in regulations such as this can have far reaching impacts in the comparability of historical results with both intended and unintended consequences. Additionally, if the changes in regulation were successful in improving athlete safety, does this correspond with prolonged athletic competitiveness and longevity given the reduction in injuries or alter our ability to compare medal achievement across time different regulation periods? We believe the answer to this question, similarly to the questions of historical competitiveness, are central to fully understanding the historical relationship between age and medal achievement. With the goals of analyzing both of these additional features in mind, it was first necessary to find a suitable indirect way of modeling both as neither are directly represented in our data set by a single or trivial combination of existing features. Unlike features discussed in subsequent sections, we strongly believe that the features included in the original data set when paired with additional research, a consistent methodology, and feature engineering would be sufficient to accurately model these two factors. The regulation feature was fairly straight forward to construct from both and implementation and methodology perspective. After researching the regulation history of alpine skiing we were able identify specific points in time where these changes took effect. Then, we were able to construct indicator variables according to filtering competition dates according to these identified breakpoints. In this way, each competition was categorized according to a given "regulation regime". In order to reduce the potential introduction of bias in our

subjective determination of "major regulations" we attempted to consult existing academic research on the impact of regulation changes in alpine skiing which is interestingly a well researched topic in both the scientific and medical research communities. However, majority of all research focused on more recent events (2000 onward) for which vastly more athlete data is available. This limitation led us only to consider regulatory changes within this frame of reference. We identified the following years for investigation: 2002/2003 season, 2007/2008 season, 2012/2013 season, & the 2017/2018 season. Existing research on these regulations either focused on the difference in competitive results given a set of regulations (reduction/increase in times) while other papers focused on the impact of injury mitigation. From a competitive perspective, research asserted that regulation changes in the 2007/2008 and 2012/2013 season significantly impacted overall times for certain events while also demonstrating contrasting impacts on the changes for men vs. women [16]. Although not directly relevant to our question of the impact of regulations on age (given we hadn't planned to make this comparison), it became important information in the construction of our competition competitiveness variable as this research clearly demonstrates times alone are not comparable overtime in a given alpine event. From an injury and safety prospective, research on the impacts of regulations for the 2007/2008, 2012/2013, & the 2017/2018 seasons were mixed. Some papers asserted that these season each represented moderate reduction in overall injury [6], while other research claimed no significant change [12]. It is notable however that Platzer's research on the 2002/2003 season (the only study of its kind found to cover this time-period) asserted that this regulation change was associated with a spike in injuries after implementation [12]. The author gives some historical context to this observation noting that the regulations exacted unintended consequences.

"Skiers who used skis that were longer than the minimum length before the equipment regulation may have switched to the new minimum length (which could be shorter). Either way, the results showed that the incidence for TOT_{SIE} [total injuries] was increased after the 2003 equipment regulation." [12]

In light of these research findings, we deemed it only necessary to inspect the regulation changes implemented in the 2002/2003 season. Many of the insights gleaned from working on the methodology for our regulation factor significantly aided our process of creating a suitable framework for measuring the field competitiveness feature. Given this knowledge, we decided that the best way forward would be to construct and index of competitive difficulty that would rank each individual historic competition by difficulty. Furthermore, the summation of an individual skier's difficulty scores multiplied by their scaled final position ranking in the competition would allow us an additional avenue to directly compare individual skier strength in a given event and gender category. This "difficulty" metric need to be hand engineered and although the feature's construction is subjective in nature we believe it to be a

reasonable approximation that provides unique value to our analysis. In computing this difficulty index we focused on three primary areas which we believe significantly contribute to a given competition's difficulty. Firstly, we considered the overall number of competitors; treating competitions with less as less difficult given the smaller competitive field. We also decided to filter out all competitions that did not meet our set number of eight minimum competitors. Secondly, we considered that races that demonstrate smaller standard deviation of time splits between the race winner and the top five finishers indicated a more competitive event. This metric functions on the principle that highly competitive competitions will be decided by thin margins. Finally, we needed to consider the number and respective world ranking of the skiers in a given competition. This metric assigned more competitive weight to competitions with a larger number and higher ranking of world ranked skiers in the given discipline. After constructing these features, we then scaled each to develop historically standardized measurements for each feature which would then be on equal scales $[0, 1]$ and thus comparable. Then, after standardisation, the scores of each metric were then summed in an equal weight fashion at the competition level to arrive at a unique competitive score for each historical competition for a given event and gender pair. In order to test the general effectiveness of our heuristic we looked at the calculated difficulty value of events we knew to be highly competitive (FIS World Cups and Olympic games) as well as less competitive competitions in order to see if the resulting model values closely resembled our expectations. Interestingly, the values for these tested events in our index generally resembled our expectations and confirmed our belief that our developed metric would suffice as a worthy approximation for our competitiveness factor. In addition to these factors, there were several other factors that we wished to study, but were unable to given our belief that it would be impractical to find suitable data or representations for within the constraints of the project.

B. Known, Measurable, but Unavailable Features

As noted previously, early on it became clear that directly comparing athlete course times was impractical, and the best way forward would be focusing on capturing competitive results via final positional rank. However, even this comparison of athletes final rank in a single event is complicated when considering the sequential nature of alpine runs which allows for natural and expected changes in both course and weather condition between athletes. This could in turn make run times ineffective comparisons. As an example, in competitions like the 100 meter race in track, each racer faces identical course and weather conditions as they all start the race at the same time making their in-event times more suited to direct comparison. Furthermore, even if we were able to control for weather conditions, sequential start position, and course design, we would still face the problem of differences in the underlying course surface due to geographical differences in local climate, competition to competition. A factor commonly

known as a "home-course-advantage". Representing the type of snow or geographic location a skier most frequently skis or prefers could thus prove an additional important factor to consider. The alpine skiing results and subsequent complaints about the use of human-made snow in the recent 2022 Beijing Winter Games although to some degree anecdotal served as the primary inspiration for this line of analysis. Finally, there are a plethora of increasingly granular potential features such as athlete physical characteristics (height, weight, injuries), training methods, frequency, and schedules would also likely be key to isolating success factors at various perspectives. Although each of these potential features demonstrates appealing potential, they are all generally impartially minute to capture or model within the constraints of this project due to unavailability, high cost, and/or limited time.

C. Known, Immeasurable, & Unavailable Features

Probably the most notable and significant feature that does not lend itself to measurement, and is inherent to the nature of alpine skiing competition is the differences in course layout and design. In continuing with our comparison to a 100 meter race, although each event has slightly different climate conditions that influence track results (primarily wind), each result is generally directly comparable from a time perspective due to the same exact distance and course design. Alpine skiing cannot be evaluated in this way as we mentioned earlier due to variance in course design and layout so times are only comparable to other competitors at the individual event level. This subtlety makes the comparison of overall final position in an event the only way to model athlete performance over both their own individual career as well as in comparison to other competitors over time as there is no record or feasible way to represent historical course structural and design elements.

D. Unknown Features

As is the case with any statistical analysis no matter how detailed and thorough, there will always be factors that escape identification by both this work and others relevant to this particular problem. Some may be identified with time while others may not. Unfortunately, the presence of these factors is not something that can be directly solved. However, I include mention of them as there are still methods of dealing with the realities of these unknown features that aim to manage rather than eliminate their potential presence and consequences. This discussion is a bit more philosophical in nature, but nevertheless is an important managerial consideration in the usage of our or any statistical model in practice. Approaching modeling with a healthy yet not excessive amount of skepticism (no matter how conclusive, definitive, or rigorous an approach may seem), limiting the degree of dependence on a single model or approach, and finally, understanding the potential downside costs a bad model could incur. With these management strategies and considerations in mind, we as researchers believe that the upside potential of well constructed models far outweighs the cost even in the presence of these

potential unknowns especially as it pertains to the task of modeling alpine Olympic medal achievement based upon age.

Now that we had identified all other secondary lines of analysis needed to validate any future conclusions about age and medal achievement, we began to grapple with the question of how exactly to formulate an experiment to probe the relation of age and medal achievement as well as our other factors. Initially, we had decided that the best way to model the relation would be from the event & gender level whereby we would look at how athlete age and medal achievement related through various modeling frameworks. However, after some insightful suggestions provided by Elliot, our USOPC project leader, we decided that the relation between age and medal achievement may be more comprehensively modeled by looking at additional perspectives by which to relate medal achievement and age. This key insight led to our decision to analyze the relation via a three pronged approach which modeled age and at the medal achievement from different perspectives namely, individual level, event level, and Country/Olympics level. By modeling the relation uniquely to each respective viewpoint, we would be able to explore the different methods and scales at which medal achievement and age are related yielding a more complete and comprehensive analysis. With this general methodology established, we began preparing our data.

VI. DATA

The primary data set used for our analysis was provided by the USOPC in csv format. The data set itself although proprietary can be generally described as historic international competition results, standings, and records for various Olympic sports. Due to the sparse nature of Olympic alpine competitions, majority of the event data utilized in this project was provided by FIS skiing competitions and world cup rankings not just Olympic competitions. The original data set itself consisted of 300K rows and 37 columns so the size was neither prohibitively large or small for our modeling and visualization tasks. Majority of the data set focused on "Elite" level competitions, but also tracked some activity at the youth and junior levels respectively.

A. Data Cleaning

Although at first evaluation the data appeared to be generally very clean and well structured for analysis, it quickly became clear while conducting some exploratory data analysis that there were several data cleaning problems that needed to be addressed. Firstly, several variables were recast to the correct data types while other that were previously stored as strings we converted to more memory efficient encoding. Several of the original columns were dropped either because they were deemed irrelevant to our analysis or because they duplicated other existing columns. Next, we spent a significant amount analyzing the unique values of each data column in order to identify if certain encoding represented potential for consolidation. Particular instances of this were the various different, but from our standpoint equivalent encoding for

DNF, DNS, DQ, and other less occurring values. All of these values were consolidated after verifying it would be suitable in our analysis to do so. Additional data cleaning challenges included the standardization and transformation of non-conforming Unicode characters, duplication of rows, and uniform handling and encoding of missing data. In terms of missing data, we identified that 17% of all athletes were missing identifiable ages/ dates of birth. Such a large quantity of missing data in a column that was essential to even approaching our modeling task proved a significant obstacle.

B. Data Augmentation

As briefly mentioned in the cleaning phase of the project, we noticed that there were a concerning number of all together missing entries for athlete ages. Given that having a large sample of highly reliable age data was central to conducting any modeling task for this project, we decided to explore alternative data sources to augment our existing data source. After extensive searches we were unable to find any useful direct data sets. However, we were able to identify the FIS website as a potential target for data scraping. The FIS website maintains probably one of the most up-to-date and detailed repositories of alpine skiing events available. Since the FIS does not support direct data downloads or queries, it quickly became clear that in order to get the data we needed we would have to develop and implement a data scraping tool that would utilize the FIS athlete search function to match known missing athletes to their corresponding entry (if any) in the FIS database [3]. Although simple in principle, this web scraper quickly became a modeling effort itself. Once it became clear that its development and usage would not only improve the quality of our modeling and analysis, but would also serve as a useful tool to the USOPC in future use cases we decided to devote a reasonable amount of time to its development. The resulting tool which we refer to as an Entity Recognition tool is composed of two parts a "generator" [Figure 25] which is responsible for generating alternative representations, combinations, encoding, and abbreviations for a given athletes input full name. The reason it is important for us to create alternative representation for a given name is so that we can search a wider scope of potential matches. Given inherent limitations in the FIS's search function we often needed to try many different representations of an athletes name to find the proper match. The second part of the tool known as the "discriminator" [Figure 27] takes the various names and aliases provided by the generator and finds general matching scraping results for each given entry. After compiling scraping results for each unique alias provided by the generator, the discriminator would then compare the original athlete name and descriptive information initially provided to the generator against the scraping results. In order to provide sufficient accuracy in matching, the discriminator first ensured that the given athlete we were looking for matched all scraped entries with respect to Olympic country, gender, and past disciplines competed in. After filtering out all scraping results that violated these constraints, it compared the similarity of

the known and scraped first, last, and if present middle names both independently and as a full name. The similarity of this comparison was measured by Levenshtein distance a popular string similarity metric that determines semblance via the number of character edits required to transform one string into the other [7]. After applying this operator to each of the names we could rank the scraping results and select the entry with the highest similarity score. This selected entry which if it contained a valid birth date or age for the athlete would be used to replace our missing values. After some tuning of the minimum similarity threshold which any valid scraped entry would be required to meet, we observed a high degree of success filling 82% of all missing entries conduction of 25,000 individual queries over a 24 hour window (to minimize server load). Furthermore, we conducted randomized testing on a subset of the total scraping results to verify the integrity of matches, and observed an error rate of 6% which we felt was a very sufficient level of accuracy. The data scraper although custom made for the FIS use case could be widely deployed for other Olympic disciplines and their respective governing bodies which typically maintain similar repositories to the FIS.

C. Data Reconciliation

After utilizing our data scraper to pull the missing athlete entries we identified during our data cleaning phase we now need to merge them into our original data set. During the process of this merge, which we expected to be a straight forward process, we kept encountering several errors. It quickly became clear that there existed several inconsistencies in the athlete age column whereby a single athlete had several possible birth dates. This inconsistency likely arose from the storage of each athlete's age at the date of a given competition instead of tracking an athlete's known birthday and deriving their age from the difference of the competition date and their birth date. In order to solve this problem we utilized two tools. Firstly we reused our data scraper, this time as a means of verifying the integrity of our existing data. By querying the FIS's birth date entries for the athletes that demonstrated an inconsistent birth date entries we were able to get additional data from an alternate source on what the actual athlete's birth date should be. In most instances, the FIS identified birth date matched the identified modal birth date from the original data in such instances the non-conforming birth date entries were replaced with the modal birth date thus creating a consistent entry for that athlete across all rows and columns. Alternatively, in cases where we were unable to find a matching athlete entry in the FIS database or the resulting birth date differed from our existing database's identified modal birth date, we took the mode of the overall entries as the reconciled athlete birthday. After applying these reconciliations to the athlete age and birth date columns we continued with the successful merge of our scraped missing athlete entries. Our final task on the issue of data reconciliation was the identification of athlete aliases and previous names. Although the Person ID column represented a unique key for each athlete, the functionality of certain user interface components of our web application

depended on athletes having both a unique and singular name and identification number otherwise they would be treated as separate entities. In order to solve this problem we scanned the data set by the unique key of Person ID to find all instances of the same ID being used with differing competitor names. After identifying these instances, we determined which of these aliases the athlete has competed under most recently. This entry was treated as the competitor's current name and was applied to all of the athlete's non-conforming entries. Additionally, the aliases that were not the most recent name the athlete competed under were preserved in a new field which listed all aliases a given athlete has competed under previously.

D. Feature Engineering

After the data reconciliation process was completed, we began to construct several additional features that we had deemed necessary to approach our modeling task. The first two of these features, regulation regime and competition strength were discussed in previously in the methodology section. From an implementation standpoint the regulation regime variable was simply an indicator term that represented if a competition was before or after our 2002/2003 season regulation change. In addition to these features, we also created several others to aid in the modeling task. Firstly, we created the "Adjusted Rank" variable. The term would serve to counter some of the potential difficulties associated with DNF values. If we removed all athlete entries with the results of "DNF" from our analysis it would likely skew our results and remove a large portion of important data. Instead of dropping these rows, the adjusted rank column was formed which replaced all DNF finishes with the a rank score one lower than the maximum of the competition's previously lowest awarded positional rank or one greater than the total number of non-DNF competitors. The advantages to this metric over the standard rank metric were crucial in our decision to model age directly against adjusted rank as opposed to the standard rank feature. Secondly, we created the scaled rank variable which is the result of min-max scaling [Equation 1] our adjusted rank variable for each competition. This variable scaling would be an important aspect of our approach to modeling individual athlete peak rank performance as it adjusts for the number of participants. As briefly mentioned in the reconciliation section, it was necessary for us to refactor the age variable into a birth date in order to ensure data integrity. After creating this birth date variable we then had to convert it to a new representation of athlete age at each given competition which we called Athlete Age Days Derived. For convenience purposes, we also created a variable called Event Hash which is the result of applying a hashing function to each row's competition id, event id, location, and date which allows the new variable to function as a unique key for a given unique competition, event, and gender triple. This variable was also crucial to acting as a foreign key connecting our separately stored competition difficulty rating metrics. Next, we created a variable known as medal score which was applied only to Olympic level competitions. This variable converts gold, silver, and bronze medal outcomes into

the following point values 6.4, 2.4, and 1. This metric was created in order to award more weight to the achievement of higher quality medals for later use in our medal forecasting models. The determination of these point assignments are far from arbitrary, and are based upon work from Sebastian Sitarz. In his research Sebastian makes the cases for two central constraints that any point system representing medals must include:

- 1) The points assigned to higher quality medals must be higher
- 2) The marginal points assigned to higher quality medals must be larger for higher quality medals than the point differential between lower quality medals

With these two constraints and the application of some rigorous mathematics spanning set theory, high dimensional geometry, and constrained optimization all beyond the scope of this paper Sitarz arrives at these stated point values for each medal as being the optimal representation of their comparative values. This system as noted in the paper is extensible to any n positional scored competition, but given our task was to focus solely on medal achievement, we did not extend this points system beyond bronze [15]. The last engineered feature we will cover is the status variable which is intended to reflect if a given athlete is retired or not. To form this variable we needed to develop a methodology for what we considered to be an "active" alpine skier, which would also become an important component to our determination of potential athletes for a given country's Olympic pipeline in a given year and event. Due to the high level of injuries and sparsity of competitions in alpine skiing, we needed to allow a fairly large window of time so as not to mistakenly rule out potential competitors. After looking at the careers of several athletes, typical time to recovery for common alpine injuries, and testing observed mis-classification rates for different thresholds among known retired athletes. We settled on a window of 1.5 years. If an athlete had not competed in any event in within this time frame they would be considered retired from the sport. With all of our data cleaned, reconciled, and augmented, we were able to finally begin the modeling portion of the project.

VII. MODELING

As previously mentioned in our methodology section, in order to effectively model age and medal achievement, it is necessary for us to evaluate the relation from multiple unique perspectives. In the following sections, we individually discuss the modeling approaches and corresponding findings for each unique perspective. Additionally, we provide analysis into one of our secondary factors of regulation and a discussion of other unsuccessful modeling approaches we attempted.

A. Athlete Level Analysis

In order to comprehensively analyze the relationship between age and medal achievement at the individual level we developed three separate modeling approaches. Our first approach centered around the fitting of splines and polynomial regression models to individual athlete career trajectories in

a given event. We had intended to utilize the smooth representations of these resulting athlete performance curves to better understanding the aging process' impact on performance through identification of peak age at the curve's maximal value as well as analyzing changes in slope between pre peak and post peak ages. However, the results of this modeling approach were generally unsuccessful, and as a result will be further detailed in subsequent sections. Our second and third modeling approaches took a very different angle on quantifying this relation. Instead of concerning ourselves with the entirety of an athletes career in a sport, instead we decided to explore selecting only the point at which they achieved their highest positional ranking. This corresponding age value associated with their peak performance could then be identified for each individual athlete for all events in which they have competed. These resulting ages thus individually represent the age at which each athlete achieve their peak performance in a given sport. In this representation, our age data is a prime candidate for non-parametric distributional analysis. Although histograms are simple and effective at demonstrating a given distributional structure of a set of observations, they rely heavily upon initial choices of bin size and aren't meant to model continuous variables such as age without additional binning or threshold. For this reason we look to the histogram's continuous variant Kernel Density Estimation (KDE) [13]. KDE loosely functions by taking a set of data points and fits a given distribution covering a certain width around that point. After placing these distributional kernels at each point according to the given kernels specified bandwidth, the distributions are aggregated to form a single continuous distribution representative of our underlying data. For the purposes of our application, we will utilize the continuous age data for when athletes achieve their peak ranking and world ranking respectively and fit KDE models to each. The resulting fit models will be distributions of these variables that we can directly sample from. By sampling these distributions and plotting their structure we can gain insights into the overall shape, modality, and strength of the relation between age and the peak of alpine athletic performance. Additionally, because we can sample these distributions, after identifying the distributional mode which will be our age of peak athletic achievement within the distribution, we can integrate left and right of our distributional mode with respect to the sorted samples of the distribution and arrive at a general peak age range. We will apply this very methodology for each of our KDE distributions under the assumption that each age category should be of equal 1/3 size. With that assumption in mind, we can integrate 1/6 of the area under the immediate left and right of the distributional mode to get the lower and upper bounds of our peak age range. This would allow us to directly identify and compare peak age ranges across both the world ranking and individual peak rank perspective, but also at the gender, and event level. In proceeding with our KDE modeling of our peak world rank achievement and peak rank achievement distributions we will cross-validate optimal kernels and bandwidth for each over a wide range

of values with 5 folds. The cross-validation process will be based upon the maximum likelihood metric [11] where by our fit models generate samples which are compared to with the probability that these samples would naturally emerge from our original distribution. The models which generate samples similar to those expected from the original data will be chosen as our optimal models [Figure 5] [Figure 6]. Once the optimal hyper-parameters are identified we then refitted the model with these parameters on full training data set. We then conducted this process for both the individual rank and world rank distributions. We then integrated our optimal KDE curves to achieve optimal peak age ranges for each event gender pair [Figure 1] [Figure 2] [Figure 3] [Figure 4]. We believe that this process created a clear insight into optimal age ranges by sport for which medal achievement potential is maximized.

B. Event Level Analysis

After conducting the individual level analysis we decided to take a similar approach to the event level modeling of age and medal achievement by leveraging the strength of kernel density estimation methods once more. One key divergence between the individual level and event level analysis was that from this higher perspective we had a benchmark to model against. Unlike the individual case which relied solely on peak rank/ world ranking for each athlete, at the event level we have a distribution of age according to various final positions not just the limiting sample of best positions achieved. The ability to then use this benchmark distribution of the age of all competitors in a given event/gender pair at all placement level to serve as a comparison the distribution of the top medal achieving athletes by age. If we are able to identify a distributional difference between both of these models then it would suggest that within that given event and gender, age plays an important role in determining the distribution of medal outcomes. however, if the distributions were sufficiently similar it would make the case that their is not a difference in the observed ages of those how achieve top positions and those who do not. With this general idea in mind, we began to individually fit kernel density models for each event gender pair on both the full distribution of athletes competing and the distribution of only the top 3 placing athletes [Figure 7]. Each of these KDE models underwent their own hyper-parameter tuning. Similar to the individual case, we utilized various kernel and bandwidth values for each and determined their optimal hyper parameters via the maximum likelihood metric [Figure 8] [Figure 9]. After the optimal hyper-parameters for each two models were identified, we applied the Two-Sample Kolmogorov-Smirnov (K/S) Test [14] to determine if the two fitted distributions were sufficiently different. The two sample K/S test is a statistical test used test the following hypothesis:

H_0 : The Two Samples Come From the Same (Unspecified) Distribution

H_a : The Two Samples Come From Different Distributions

By calculating the K/S statistic for each of these paired distributions, and then converting the statistic to a p-value,

we would be able to make a definitive conclusion about the acceptance or rejection of the above null hypothesis. If the p-value for a given event gender pair is sufficiently small (consistency we use the threshold of $\alpha < 0.025$) then we would be able to conclude that there are significant differences in the ages of alpine athletes who achieve high finishing positions and those that do not. This statement thus implies that age plays a significant factor in medal achievement in that given event gender pair. After running each individual event gender pair through this modeling framework (as well as combinations of the two) we arrived at the unanimous conclusion that in each case we tested the distribution of all competitor ages and top placing competitor ages was significantly statistically different. It is rare to in this type of analysis to see such a resounding agreement in individual cases, so when you do see this type of agreement it is fairly definitive it its interpretation. In addition to testing the K/S statistic for each of these event gender pairs we also recorded each top medal athlete distributions peak age boundaries for later visualization. The identification of these peak age bounds was similar to the process described in the individual modeling section utilising the same $\pm 1/6$ area under the curve from the distributional mode [Figure 10] [Figure 11]. Unlike our previously identified age ranges for the individual analysis case, due to our proof of age significance in the event level case we have additional justification that the peak ages identified represent unique and optimal boundaries on which to categorize athletes. In later pipeline and country level analysis it is these identified age ranges that we will utilize to effectively determine a given country's number of athletes in each peak age category. When paired with our individual peak analysis, the event level analysis makes an extremely strong case for the presence of a sizable and identifiable relation between peak age and medal achievement in alpine skiing across all events and genders.

C. Regulation Analysis

In a similar approach to our event level KDE model of the distributions of top vs. all athletes in a given discipline, we decided to measure the impacts and effects of regulations by applying KDE to the data before and after regulation implementation separately. Each model would have its own optimal hyper-parameter for bandwidth and kernel which represented the highest maximum likelihood of generating the underlying data. After fitting each model, sampling, scoring, and plotting their respective distributions, we would calculate the two sampled K/S statistic. If the test statistic reached a significance threshold $\alpha < .025$ the null hypothesis which states the distributions are non-distinct would be rejected, and we would accept the alternative hypothesis that the distributions are sufficiently different. If the distributions were observed to be sufficiently similar, we would infer that the implementation of regulations had no observed effect on the underlying distribution. Alternatively, if deemed sufficiently different it would indicate the observed difference in distributional parameters suggests the change in regulation corresponded with a change

in the other underlying variable age. In keeping with our method logy, it would be necessary to model the impact of regulations with respect to both age and medal achievement in order to identify which side of the relation (if any) the regulation variable impacts. In testing the impact of regulation with respect to age, we compared the fitted KDE models of the distribution of all elite level athletic competitors before and after the implementation of regulations. By comparing the aggregate ages of all competitors we would be able to see if the regulations altered the age of the competitive pool at large. Interestingly, at the global level (all genders and events), gender level, and event levels we were able to identify no difference in the underlying distributions thus indicating that regulations did not significantly impact the ages of the competitive pool. We then in an identical framework compared the distributions of age of top medal achieving athletes (positions 1-3) before and after these new regulations. Similar to the previous result at the global level we were unable to identify any significant differences in the two distributions based on our resulting K/S tests. However, at the event & gender levels Super G and Downhill for both men's and women's events displayed a significant difference before and after regulations. In the case of all four we can clearly a shift in the distributional mode rightward after regulations were implemented. As an example see [Figure 12] which shows that after the implementation of regulations the observed distribution of top medal achieving age for Men's downhill shifts to a higher value indicating in the post-regulation environment older athletes saw a boost in medal achievement. This observation of rightward modal shift was consistent across these four events demonstrating evidence that the regulations most strongly impacted downhill and Super-G events. These results might also suggest that the improved safety standards worked in achieving their mission of prolonging athletic life in these two most dangerous of alpine disciplines. However, despite these effects given that they are isolated to only these few instances and at a relatively slight to moderate scale, we believe the effect measured is not significantly large or prevalent enough to meaningfully confound or augment our prior conclusions on the relationship between age and medal achievement.

D. Medal Prediction

One of the other major questions we hoped to answer though the identification of peak age thresholds was whether or not they would hold any predictive value in medal achievement. We believe that answering this question on an individual competition basis would be extremely difficult if not impossible. This belief is based upon the random nature of single event's outcome/prediction, the small amount of data each model would have available, and the sparsity of Olympic competition. With this in mind, we decided to approach the problem of medal prediction would be much more suitable at the country level which allows us to avoid most of these issues due to our ability to aggregate a more sizable sample across all alpine event gender pairs. The model would take as input the number of identified pre-peak, peak, and post

peak athletes for all event gender pairs in a given Olympic games for each country, and would predict the observed medal score actually achieved by the country in that specific Olympic competition. This modeling framework which takes the form of a regression problem was implemented into five specific models Linear Regression, Ridge Regression, Lasso Regression, Regression Trees, & Neural Network. After preparing and formatting our data into a representation suitable for input into each model, we split the data into a training and testing set according to a random 70/30 split. Each of these five models underwent five fold hyper-parameter cross validation against the training set. Due to the large number of parameters and models we tested, we will refrain from discussing them in too much detail individually. For linear regression we varied the inclusion of intercept term and normalization of features [Figure 13]. For Ridge regression, we varied the inclusion of an intercept term, normalization of features, and value of l_2 regularization commonly referred to as α [Figure 14]. For Lasso regression, we again varied normalization, intercept, and l_1 regularization also known as α [Figure 15]. For regression trees, we varied maximum tree depth and minimum leaf size [Figure 16]. Finally, in our last modeling approach we utilized 5 differing architectures of neural network each with a varying number of hidden layers and neurons [Figure 17]. Since hyper-parameter grid search methods are infeasible for neural network models our architectural choices were more subjective, but were generally chosen in increasing number of dense layers [0 – 4] and number of neurons [32, 64, 128, 256] sequentially from models 0-4 respectively with only model 4 having a unique fifth dropout layer. Each model utilized the same activation. After identifying our optimal hyper-parameter values for each class of model, we retrained each class of model with its optimal parameters on the training data set. Each of these resulting fitted models were then used to generate predictions for both the training and test data sets, and pickled for later use/reuse. These resulting model predictions were then compared against their respective true medal scores utilizing mean squared error as our evaluation metric the results of which are summarized in [Figure 18]. As demonstrated in the table, we can see that all models demonstrated relatively similar performance with our Neural Network model just edging out the advantage against the test set. However, given the strong similarity in overall model fit, and taking into consideration the disadvantageous interpretational nature of neural networks, we decided that the Ridge regression model would serve as our preferred model going forward. Although the ridge model had a slightly worse MSE than the neural network model, it offers far more in terms of value given its simplicity and ease of interpretation with respect to coefficients. The ability to inspect and better understand the values of our age coefficients will offer significant advantage in terms of providing an answer to the question of the peak age variable's significance. As we can see from the plot of all our regression model coefficients in [Figure 13]. It is the case in all three model classes the peak age coefficient takes the large positive value. Combining this with our ANOVA

outputs [Figure 18] which allow us to quantify the statistical significance of each feature, we can see that under each framework the peak age variable is statistically significant at $\alpha < 0.025$. Furthermore, from an overall fit perspective we can see that our ridge model demonstrates the highest R^2 value among its regression peers which shows that with these three variables our model was able to explain 31.24% of the variation present among medal scores. Although these metrics are not solely definitive in determining whether a model is suitable or not, they do give us sufficient evidence to conclude that the model is generally speaking useful in this prediction problem. With these strengths in mind it is important to note some of the underlying issues with our modeling approach. Given that these models take as input individual observations from various different countries and Olympic games without a variable representing either context, the model will return predictions that when taken as a groups (for example if returned over all counties for a single Olympic games) are not observably possible as the sum of their medal scores will typically be much larger than the known sum of medal scores across all countries in a given Olympic games. For this reason, each of our models are inherently removed from reality. There exists no exact remedy for this problem as including variables that add context to country and Olympic games would lead to over fitting while still not guaranteeing realistic medals scores either. Multiple input multiple output and batched regression approaches were considered, but ultimately not pursued due to additional complications and limitations of these frameworks the most central was the inability to generalize beyond a given batch which would prevent any such forecasting of Milan 2026. The only other solution we tested involved the re-scaling of predictions to a $[0,1]$ scale using mix-max feature scaling [Equation 1] applied at the Olympic games level. Then we would re-scale these resulting scores by the known observed sum of medal score points for that given Olympic games. This forced the predicted medal scores for each Olympic games to match the observed sum of medal points actually observed. This trick allowed our model to now match reality more strongly, but soon we noticed that in comparing the MSE of our new predicted values to the known target medal scores, our predictive performance dropped significantly nearly doubling our observed Mean squared error on both the test and training sets when compared to the original un-scaled model predictions. These poor results led us to return to our original un-scaled models and predictions, and present them as the representation of our medal score modeling efforts. Despite the underlying issues with these models, they still demonstrate significant utility to analysts and users aware of its limitations. One novel way we discovered to reduce this effect was to transform the predicted medal scores into a ranking system and then using these transformed values to compare to the observed rankings of realized Olympic medal results. Formulated in this fashion the model no longer suffers from the problem of medal score constraints, as it only focuses on relative ordering which although a lossy representation, better reflects a great way in which the model should be used.

Namely to understand generally how country can expect to perform in a given Olympic games given where their pipeline stands at a given moment in time. Collectively, we believe the results of our medal prediction models to be sufficient to answer several of the USOPC's central questions. Firstly, our models and their corresponding visualization within the accompanying web application successfully model Team USA's relative pipeline strength overtime and with respect to other countries in all provided Olympic games [Figure 22] [Figure 24]. Additionally the significance of the peak age variable when paired with the overall model in general clearly show that peak age can serve as a key predictor to medal prediction explaining nearly a third of the variation among medal scores achieved. Finally, given our specific design choices in our modeling methodology, each model can be used to predict forward looking Olympic medal results. Although only time will tell if our predictions for Team USA's performance in Milan are accurate, we believe that our approach gives us the best chance for success in this prediction task.

E. Pipeline Analysis

One of the primary goals of the USOPC for the outset of this project was to better understand how Team USA was positioned in terms of potential future athletes with respect to other Olympic countries. Understanding and monitoring the composition of an athletic pipeline is key to ensuring that Team USA will be able to effectively compete in as many disciplines and events as possible thus maximizing their probability for medal achievement. Additionally, monitoring the health of athletic pipelines allows managers to make the proper investments in junior and youth programs according and ahead of their future needs. This is especially crucial given the large timelines that the USOPC operates over. With these goals in mind, we set out to create a metric that would successfully model pipeline health with respect to athlete age. We decided to refrain from incorporating FIS world ranking information within our metric for athlete pipeline strength for several reasons. Firstly, due to the high weight the FIS rankings place on Olympic events, world ranks fluctuate highly after a given winter games. Additionally, given the structure of our data set there were several edge cases whereby an Olympic world ranking reading would overlap with or follow an Olympic running. Although isolated in effect, if not rectified it would cause potential label leakage for those given observations as the world ranking would strongly mirror Olympic results thus leaking information about our true label into our explanatory variables. Secondly, after further researching the method in which the FIS calculates world ranking metrics, we began to have some concerns about its relevance to Olympic performance. Given the sparse nature of Olympic competitions, in order for an athlete to realize medal ambitions they must demonstrate prowess in both consistency and speed. Contrary to this dual objective, the FIS rating are calculated according to an average of a given athlete's two best runs in an event over their evaluation window. The calculation of the metric in this fashion thus encourages all

or nothing performance not consistency. These concerns were further validated by existing academic research and criticism about the ranking metrics [8]. It is this difference in objective, along with our label leakage concerns, and our inability to use any resulting model including world ranking in forward looking predictions (given we won't know world rankings close to Milan 2026 for many years) that led us to decide not to include this feature in our pipeline strength analysis. Instead the metric of pipeline strength would solely be a product of the number of potentially competitive athletes in each respective event/gender's corresponding peak age categories at the time of Olympic competition. Each country's age category counts were then weighted by their respective scaled regression coefficients as determined in our best ridge regression model from the medal score modeling task. This weighting scheme allowed us to weight the strength contribution of each distinct age category in an unbiased and optimal way. This weighting is necessary to aggregate these individual category scores into a single metric of a country's pipeline strength in a given Olympic season. Although this metric could be furthered by the inclusion of other potential useful features, in keeping with our stated goals of modeling pipelines with respect to age we decided to keep it solely focused on age. We did however later decide to extend these pipeline strength metrics into a rating system for both pre-Olympic rating for each country in alpine as a whole as well as ranking with respect to each given event gender double that the country will potentially compete in. In terms of relating these approach to the central questions posed by the USOPC, this metric (when visualized in our web application) will provide a clear indication of the health of Team USA's alpine skiing pipeline health and how it has evolved over time [Figure 22] [Figure 23]. Although this rating and scoring system for pipeline strength is not a flawless system that depends on several underlying assumptions especially the peak age ranges for each gender and sport identified in the event level kde analysis, we believe it to be one of the only reasonable approaches at answering this question solely with respect to the age variable as posed in our central questions. This view is further supported by our analysis of the correlation between our engineered pipeline strength index and the paired Olympic medal results which come in at $\rho = .67$. Additionally, the single variable regression of our pipeline strength feature on paired Olympic medal results similarly demonstrates the feature is significant at the $\alpha < 0.025$ level in explaining 47% of the variance (²) found in observed Olympic medal results. These results also suggest the wider conclusion of age's significance as a feature in predicting medal achievement given that the pipeline strength index is a metric solely conveying information about athlete ages. Relating these insight back to our central questions posed by the USOPC we can confidently say that a country's count of athletes competing at an Olympic games while in their peak age ranges in their respective sports show some capability of predicting medal success. Additionally, this results also suggest that the number of pre-peak athletes within a countries pipeline can be used to effectively forecast future games such

as Milan in 2026.

F. Approaches that Didn't Succeed

In our search to arriving at our production models, there were several approaches and methodologies that although initially appealing and applicable to our task just did not produce insightful results. Firstly, we spent a considerable amount of time looking at the application of time-series clustering techniques. It was our hypothesis that if we were able to compare individual athlete career trajectories (the athlete's observed competition ranks over time for a given event) and group them according to their similarity that we would be able to identify natural novel clusters of similar trajectories. The existence of distinctly clustered career trajectories would indicate that there are reasonable archetype trajectories for different categories of skiers. Such an insight would've indicated that we needed to analyze medal achievement and age in the context of these distinct athlete groupings as their given cluster could alter the underlying relationship. In the pursuit of this analysis, we applied various modeling approaches including k-means with dynamic time warping [10] and hierarchical clustering methods [9]. Each modeling approach was applied to all event gender subset individually removing career trajectories that contained fewer than 15 competitive events. However, in each case we received very poor results with silhouette scores which are the primary metric used to quantify the quality of cluster assignments. The metric which ranges from $[-1, 1]$ was not significantly different from zero indicating that clusters were entirely arbitrary with no significant separation between them. We also attempted the same modeling approach utilizing individually cross-validated best fitting B-splines [2] to represent a smoothed version of each athletes given career trajectory. We thought that this smoothed representation would allow the algorithms to more effectively cluster the series however our results were nearly identical to the original representations. Although a poor result in terms these given models, the knowledge that the trajectories of alpine skier careers were not reasonably clustered gave us confidence in our selected individual athlete modeling methodologies which did not rely on the separate modeling of additional sub-groups. Our second modeling approach that didn't seem to demonstrate much value in our project was the utilization of Facebook robust word embedding [4] within our data scraper. Originally, our data scraper identified best matching athlete entries from the scraping results utilizing a similarity metric derived from aggregating the ranked edit distance (derived from the Levenshtein distance operator) and the ranked cosine similarity (derived from the robust Facebook word embedding) of the the target and scraped names [Figure 26]. Our intention was to leverage the more general ability of this pre-trained word embedding to capture similarity and representation of nicknames that may go un-captured by the ranked edit distance metric alone. However, after inspecting the scraping results, matches, and score contributions it quickly became clear that the inclusion of the word embedding offered no additional value from a prediction standpoint. In order to verify that

it was the Levenshtein metric alone was responsible for the high performance of the scraper we separated and tested each independently the results of which clearly showed that the Facebook robust embedding alone was not sufficient to match our scraped entries to true labels. It is this reason that our final scraper only relied upon Levenshtein distance. Our third unsuccessful modeling effort centered around our initial approach to modeling medal achievement and age at the individual level. Our first approach in modeling the relation centered on the idea of aging curves [1]. Generally speaking, aging curves describe the scientifically observed variations in certain tasks over time which usually take the shape of an concave parabola indicative of improvement in task performance until a peak performance is reached following which there is a subsequent decline. Since the concept of these aging curves proved effective at modeling other physical task capabilities over time, we thought it would be a great starting point for our analysis of individual athlete performance over time. Our initial hypothesis was that similar to the prototypical aging curve, we would see an athlete's rank gradually improve, reach a peak, and then subsequently decline. The identification of such a curve and the age at which it achieved its corresponding maximum value would thus provide us with insight into if and when an athlete reaches peak performance. Additionally, the slopes of these curves could be inspected further to develop an approximate magnitude of effect associated with gain/decline of competitiveness with age by evaluating either side of the curve's maximum value. Alternatively, if our fitted line turned out to demonstrate minimal curvature, no maximal value, or was far too noisy to conclude anything we could conclude that for that given individual there is no identifiable age curve or effect. Further expanding this method, we would in-theory be able to align these fit age curves according to age then average them to arrive at a archetypal aging curve for the a given alpine discipline. With these goals in mind, we began researching ideal methods for fitting these individual career trajectories which demonstrate a-periodicity and high volatility. Rudimentary approaches such as moving averages and exponential smoothing proved unsuccessful yielding extremely noisy and non-smooth which completely undermined our ability to identify any reasonable modality or slope. Eventually, we moved to modeling methods that enforced some degree of smoothness on the overall curve namely splines. In this modeling effort, we evaluated three different flavors of spine: B-splines [2], and A-splines [5]. Each of these types of spline demonstrate unique individual qualities which distinguish them, but generally speaking are very similar. Only A-splines are uniquely distinguished by their proposed theoretical ability to optimally select the location and number of knots implicitly. However, after significant modeling of each of these three types of spline it quickly became clear that the added computational cost of calculating A-splines when compared to B-Splines was not justified by significantly better or different fit. Due to the fact that these spline models needed to be cross-validated and fit quite efficiently in order to process all individual athlete career trajectories for all unique

sports the A-spline model was dropped from our analysis and modeling effort. With our spline models narrowed, we needed to now decide on optimal hyper parameters for each method. Within the class of spine models, there exist three main hyper parameters to modify, the degree of the spline, number of knots, and location of knots. The first parameter, degree, determines the degree of polynomial used in the formation of the spline. Having a higher degree will enable a better fit (from an error perspective) as it allows more interpolation. Alternatively, a lower degree will put restrictions on the global function shape thus increasing error, but reducing over fitting. If our hypothesis about aging curves proves correct, we should observe that best fitting splines would naturally tend to be of degree 2 thus a parabolic curve. Lower degree splines would indicate constant linear increase or decrease performance with age while higher degree splines would result in very non smooth complex relations between age and athletic achievement. The second and third hyper-parameters for spline models are number of knots and location of knots. Knots play an important role in the flexibility and fit of a given spline as they constrain a spline to taking on certain values and continuity at each given placement. Increasing the number of knots similar to degree adds more flexibility in fit and vice versa. However, in addition to the number of knots the placement of these knots is also crucial. If outliers within a data set are selected as knots for a given spline the fit of the spline globally will be very poor thus it is imperative to pick points representative of your data set. Additionally, selecting knots that lie near to sudden changes in observed data patterns can also be beneficial as the curve segment prior to the knot will be able to model a different curvature or concavity than the curve segment following the knot thus allowing better model fit across distinct change points in a temporal series. In addition to these spline methods, we also created and applied polynomial regression models. Three flavors of these polynomial regression models were implemented using varying degree of polynomial from degree 2 to degree 4. For each of the corresponding models we applied 5 fold cross-validation to determine each model's optimal hyper-parameters, or in the case of the regression models optimal fitting coefficients. For our spline models, we tested degrees ranging from 1-4, for minimum knots we tested values ranging from 2 to $n/2k$ where n represents the number of data points in a given trajectory and k represents the number of cross-validation folds, and lastly since we were using k -folds cross-validation, we allowed for both random and equally spaced knot positions. Mean Squared error would be used as our evaluation metric for modeling fit. After fitting all of these various models across identified full (start - retirement) athlete career trajectories it quickly became clear that we would not see any useful results out of this modeling framework. The tabulated counts of our best fitting splines degrees were nearly uniform indicating that there was no preference towards the degree two aging curve. Additionally, the aggregated curves due to underlying differences in fit degree were extremely non-smooth, and thus useless in our identification of an aggregate age/performance trend. Generally speaking, the MSE errors

for our best fitting models were rather unsatisfactory and demonstrated that the models did not really do a great job at representing the underlying data. This is a testament to how volatile and non-smooth individual alpine skier's perform in their career and demonstrates that modeling them is incredibly difficult. Although the approach itself was unsuccessful, we did decide to retain the models and code for use within the web application [Figure 20]. As you'll notice in the individual dashboard, there is an interactive plot of your selected athlete career trajectory for the selected event. This plot allows the user to select and view various implementations of these smoothing model to see first hand how they forecast and predicted (via extrapolation) a given athletes performance. Our fourth and final modeling approach that wasn't able to generate any meaningful results was our analysis of field competitiveness overtime and its potential impacts (if any) on athlete medal achievement and age. Although we were able to successfully construct a measure to represent this factor, as referenced in our methodology section, we were unable to take away any real insights from it, which almost universally increases with time. This insight although interesting in that it show that alpine skiing is continually seeing a more and more competitive field, it also demonstrated that our metric of calculation had some very high bias toward more recent events. On further inspection, this is partially due to the weight given to number of competitors and world rank which have both expanded in size and length respectively in recent years. Although we still find the metric useful and interesting in some of our dashboard visualizations, we decided that given the bias, and no additional time to explore other approaches that we would exclude it from our analysis. Despite the lackluster success of these specific approaches, each provided us valuable insights which certainly contributed to our successful modeling approaches.

VIII. APPLICATION

In continuing with our methodology of examining the impact of age with respect to three different perspectives, we developed our web application to have three primary dashboards each of which is uniquely devoted to one of these perspectives from which to further analyze age. Each dashboard contains multiple interactive charts, figures, and tables that allow users to explore our analytical results firsthand as well as experiment on their own. Generally speaking, this application is intended to be front facing view of our project while this paper serves to add more context behind our direct analytical results and corresponding methodologies.

A. Infrastructure

From an infrastructure perspective, our application relies on several core technologies to ensure wide compatibility across devices. All of the application, modeling, and cleaning code is written in Python. Several Python modules such as Pandas, Numpy, Sklearn, and Scipy were crucial to the modeling and cleaning workflows. Tensorflow was also leveraged to construct the neural network models to efficiently

train on GPUs. Selenium was utilized in the data scraping workflows as the primary module which fetches and returns our desired results from internet sources. The most critical python module to the application would be Plotly Dash which is responsible for the hosting and serving of all our data visualizations as HTML elements from a local flask server. We had initially planned to utilize Tableau as our primarily visualization medium and had prepared an SQL database for this explicit purpose. However, later in the project it became clear that Tableau would not be able to over the interactive and modeling focused user experience we had originally hoped to deliver. For this reason we switched to using Plotly Dash as it provided us the flexibility to offer the interactive and customized components for our modeling sections we had envisioned. All of the Plotly Dash components were further styled with SCSS which compiles to standard CSS style-sheets. Tableau was also utilized as a visualization resource by embedding our generated Tableau public visualizations directly within the web application. All source control and code repository storage was managed through GitHub. Conda was utilized in maintaining the local python environment, and served as the base image of our applications docker image. The application after being fully containerized with docker was moved to a private repository on docker-hub with repository access keys and controls given the USOPC for their continued use. It was our decision to disseminate the project using docker in order to provide an operating system agnostic and simple startup experience. For detailed instructions on running the application, please see the accompanying docker instructions. On the presentation side of our project, $\text{\LaTeX}$ was utilized in the preparation of all written reports and bi-weekly updates, PowerPoint was utilized for slide presentations, & Adobe Photoshop and Draw.io were used in the creation of certain application user interface components. We hoped that our choices in underlying infrastructure would allow the application to be easily maintained and widely accessible to a wide variety of users. In the subsequent sections we will introduce, explain, and highlight various visualization and modeling capabilities our web application provides.

B. Athlete Dashboard

The Individual athlete dashboard is intend to provide users with all of the information they could possibly need in evaluating a given Alpine skiing athlete. The dashboard itself consists of eight individual interactive visualizations each depicting an important dimension of better understanding athletic performance. Starting from the upper left of the application and working down and rightward, we have the following elements [Figure 20]:

- 1) Athlete Biographical Information Section
- 2) Interactive pie chart that displays the athlete's participation by event and filterable by season
- 3) Interactive column bar (or box depending on selections) plot that display an athlete's finishes, podiums, and seasonal performance.

- 4) Scatterplot of a given athlete's competition performance and the corresponding competitive index score (both min-max scaled).
- 5) Kernel density plot of individual peak athlete rank (in given event gender)
- 6) Line chart of the selected Athlete's world ranking by event over their career
- 7) Kernel density plot of individual peak athlete world rank (in given event gender)
- 8) Career trajectory plot of the selected athlete's selected event with options to fit spline and regression methods for interpolation.

C. Event Dashboard

The event dashboard represents the primary entry point for additional modeling, analysis, and visualization of our event level modeling methodology. This dashboard allows far more in depth modeling and cross-validation of all KDE models discussed throughout our report. The dashboard itself is composed of 5 unique components [Figure 21]:

- 1) Filterable and Dynamic Violin plots utilizing each of our various KDE modeling approaches
- 2) Dynamic and filterable Kernel Density Plot that allows the plotting of probability density, cumulative density, & probability density difference for all of our discussed KDE methodologies. Also provides user functionality for custom peak age selection, kernels, and bandwidth parameters for added functionality.
- 3) In-browser cross-validation plots of the kde hyper parameter tuning process. Both update with new options selected.
- 4) Interactive tables which show each KDE method's identified peak age ranges by sport and gender.
- 5) Kernel density comparison tool whereby users can use custom input parameters to compare different KDE models and approaches.

D. Country Dashboard

The final and probably most important dashboard within our web application is the country dashboard which aims to provide users with a complete view of a given country's position in the race to Olympic gold. The dashboard itself is composed of eight individual components [Figure 22] [Figure 23] [Figure 24]:

- 1) Interactive & filterable column bar plot which shows Olympic medal counts overtime
- 2) Embedded Tableau dashboard which houses a dynamic choropleth representing all time Olympic medal count leaders at the country level
- 3) Interactive Column bar plot which shows a country pipeline age composition for a given Olympic games
- 4) Pipeline ranking table linked to the column bar chart that shows how our pipeline strength metric ranks countries
- 5) Post-hoc Olympic results table that show actual medal counts observed also linked to column bar chart selections

- 6) Country pipeline comparison tool which allows the selection and comparison of countries in a given Olympic games; includes a box-plot representing historic pipeline strength for both countries, and a table representing the differences in their pipelines by age, rank, and perceived strength.
- 7) Country pipeline drill-down tool which given a selected country creates box plot of that nations historical Olympic pipeline strength and ranking according to either gender of event; Also includes tables with all athletes by event, gender, and age that have/had the potential to compete in that given games.
- 8) Olympic Medal Score prediction plots viewable from either a country or individual games perspective for both rank and medal score objectives

The elements within this dashboard in particular allows users the capability to explore and find answers to questions concerning pipeline strength, future athletes, and Olympic competitiveness. As opposed to enumerating our findings in answering the USOPC key questions 1, 3, & 5 we instead refer to the exploration of this dashboard which allows users to answer each in a detailed and user-friendly fashion.

E. Additional Utilities

In order for our application to remain a useful tool to the USOPC it was necessary to implement functionality for updating both the application's underlying data and models. From a data perspective, we implemented an additional page within the application that allows user to upload new athlete and event data. Internally, the application first verifies the csv or xls file's schema matches the required format, concatenates the rows with the backup of our existing uncleaned data, and then applies our cleaning script to the augmented data set. After the cleaning script has finished running, the application also automatically updates the other dependent data files to reflect these changes. In addition to the manual uploading option for data augmentation, we also included several scraping scripts which can be used by the USOPC to scrape up to date information from the FIS repository. From a modeling perspective, it is natural to expect that models used over longer time horizons will eventually begin to decay as they no longer represent a reasonable fit to more recent observations. It is with this problem in mind that we also created functionality within the application to automatically retrain all our models after certain time and data addition thresholds. Collectively, we believe these added utilities provide significant value to the application in terms of usability and longevity.

IX. FUTURE WORK

Given the inherent time constraints of the practicum there were many modeling ideas and approaches that there just was not enough time to fully explore. As mentioned previously in the methodology section, there were many variables that may have significant modeling value. In future works, the aggregation and investigation of these additional features is probably the single biggest avenue for expanding our existing

work. From a modeling perspective, if given additional time we believe we would have been able to develop a more elegant solution to the medal prediction task that didn't rely upon additional post prediction processing and transformations to serve as a reliable predictor. Although there wasn't enough time to complete our research into such alternative methods, there were several promising lines of research that were never fully explored. The final aspect of future research we find especially notable is a more detailed treatment of the non elite classes of athletic competition. Due to the lack of junior and youth data, we were limited in our ability to analyze these groups. In future works finding more detailed data on these younger skiers could provide additional insights especially as we attempt to model Olympic games further in the future.

X. CONCLUDING THOUGHTS

In summary, we believe our analysis provides clear evidence that within the sporting discipline of Alpine skiing, the age of a given athlete play a significant role in Olympic medal achievement. Although the effect and range of peak age ranges by gender and specific event, the effect is notable in all avenues of our analysis. However, that being said there is also evidence that a wide variety of other latent factors have significant impact of medal achievement beyond age as we were only able to explain 31% of the variance. It is our recommendation that future studies should explore alternative factors of medal achievement. Although our efforts at Olympic medal score prediction and pipeline modeling were generally successful with reasonable out-of-sample testing results, we still believe that these models should be used with caution as due to the limited sample size available and feature set, they may not generalize well going forward.

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APPENDIX A RELEVANT EQUATIONS

Equation 1 : Min – Max Scaling
$$x_{scaled} = \frac{x - \min(x)}{\max(x) - \min(x)}$$

APPENDIX B ASSORTED FIGURES & TABLES

Figure 1. Peak Age Ranges According to Individual Peak Rank Achievement KDE (Segmented on Gender)

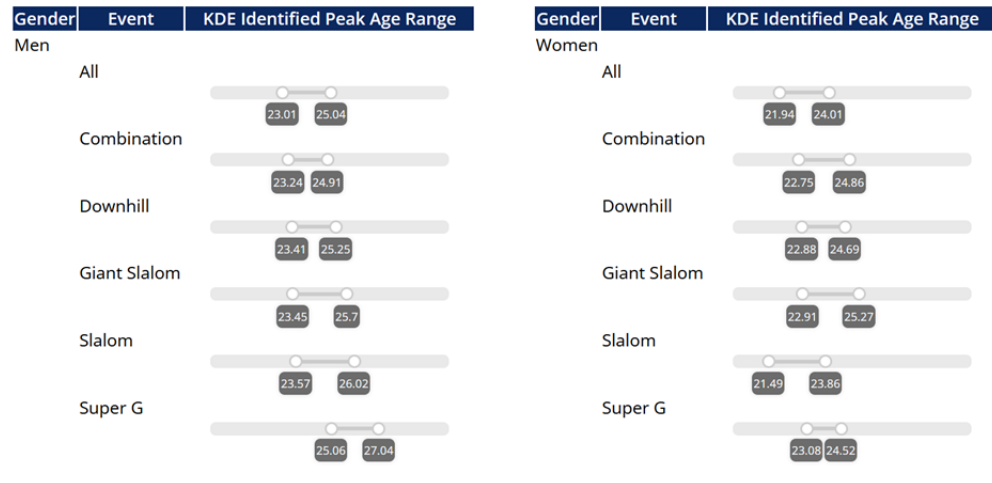


Figure 2. Peak Age Ranges According to Individual Peak Rank Achievement KDE

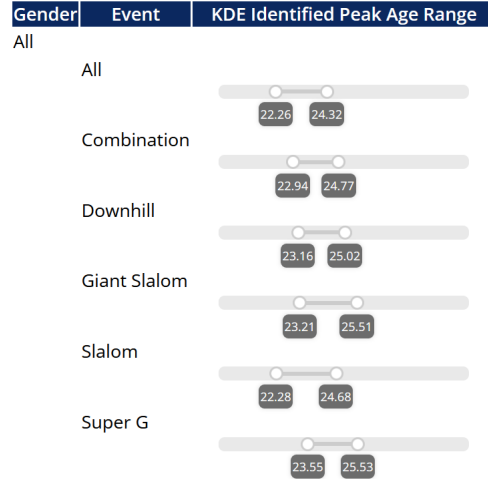


Figure 3. Peak Age Ranges According to Individual Peak World Rank Achievement KDE (Segmented on Gender)

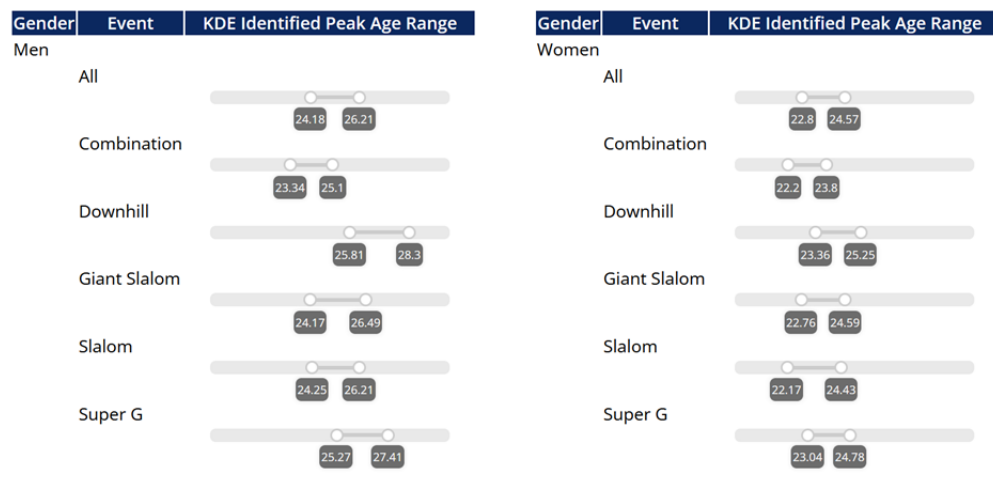


Figure 4. Peak Age Ranges According to Individual Peak World Rank Achievement KDE

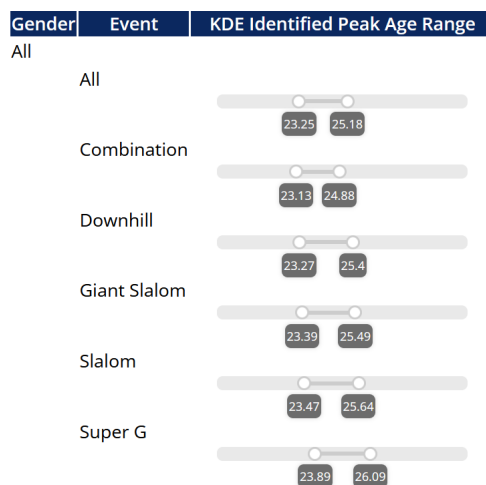


Figure 5. Individual Peak Rank Fit KDE Model for Men's Downhill Skiers

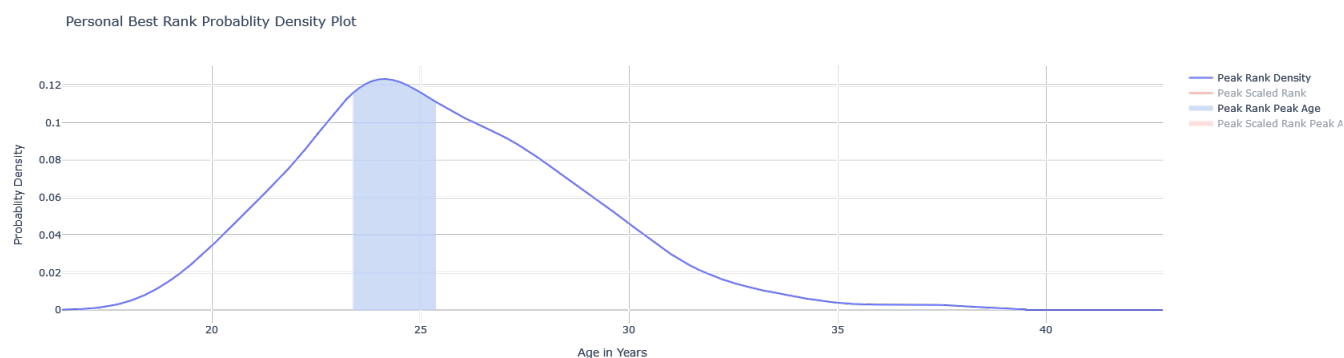


Figure 6. Individual Peak World Ranking Fit KDE Model for Men's Downhill Skiers

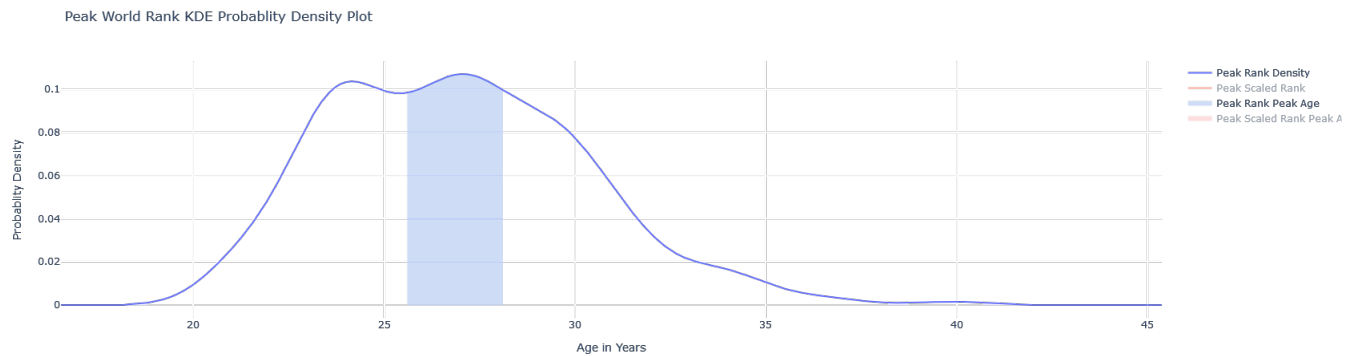


Figure 7. Event Level Fit KDE Model for Men's Downhill Skiers

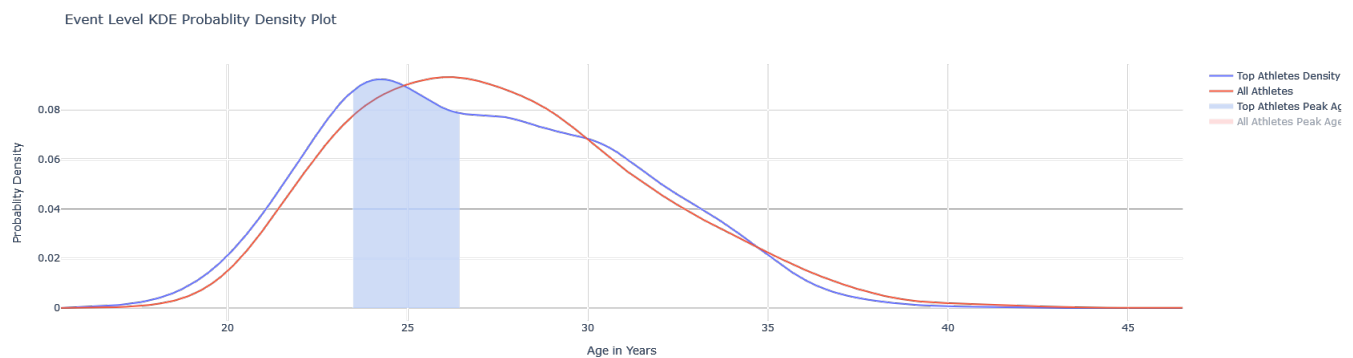


Figure 8. All Athletes Event Level KDE Cross-validation for Men's Downhill Skiers

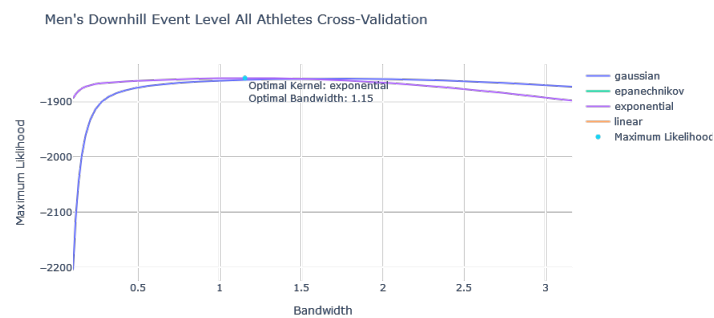


Figure 9. Top Athlete Event Level KDE Cross-validation for Men's Downhill Skiers

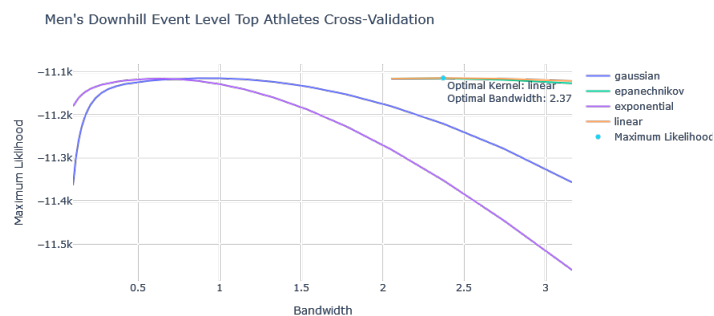


Figure 10. Peak Age Ranges According to Event level Top Achievement KDE (Segmented on Gender)



Figure 11. Peak Age Ranges According to Event level Top Achievement KDE (Segmented on Gender)

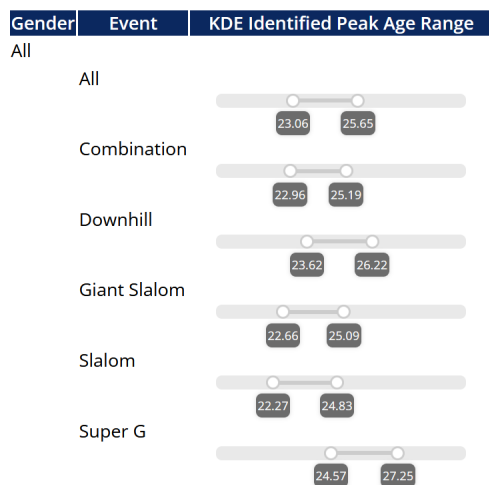


Figure 12. Fitted Regulation KDE Model for Men's Downhill Skiing

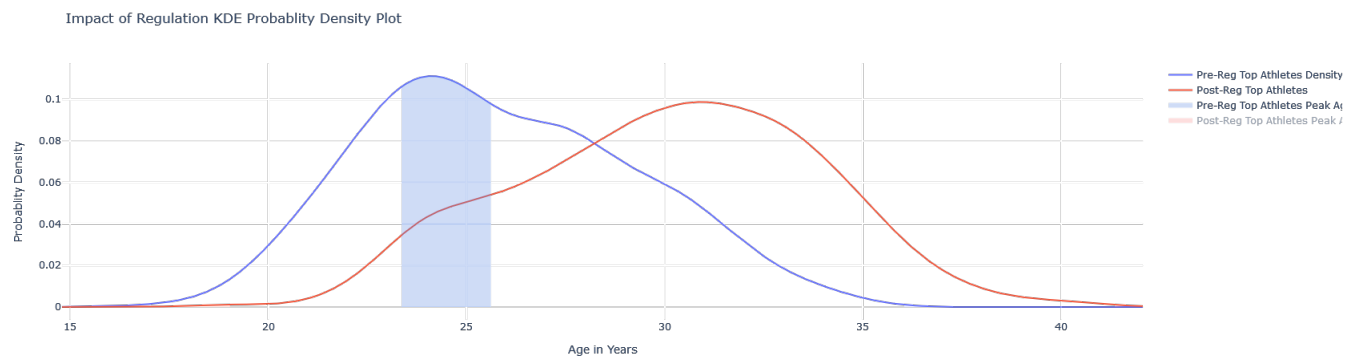


Figure 13. Medal Score Modeling: Regression Coefficient Comparison

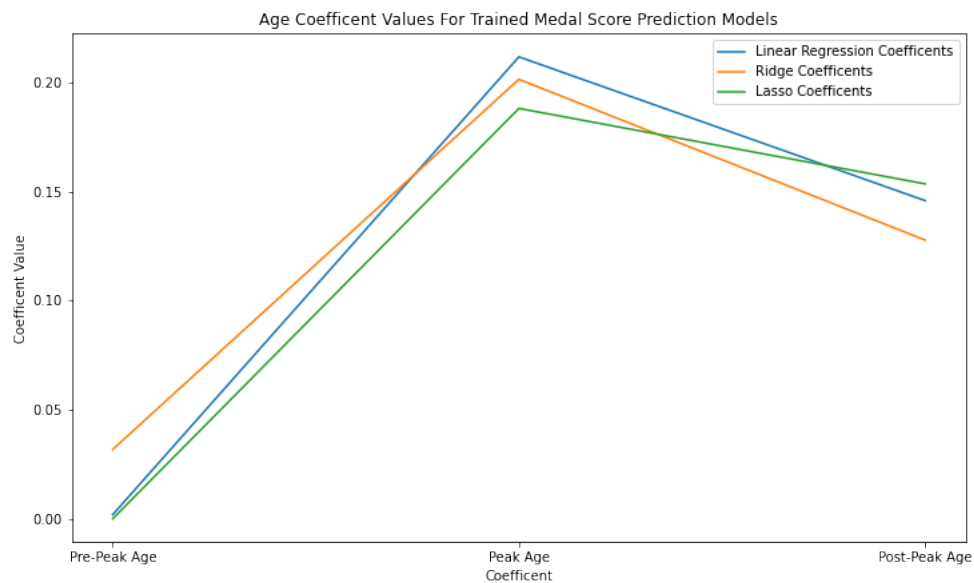


Figure 14. Medal Score Modeling: Ridge Alpha Hyper-Parameter Cross-validation

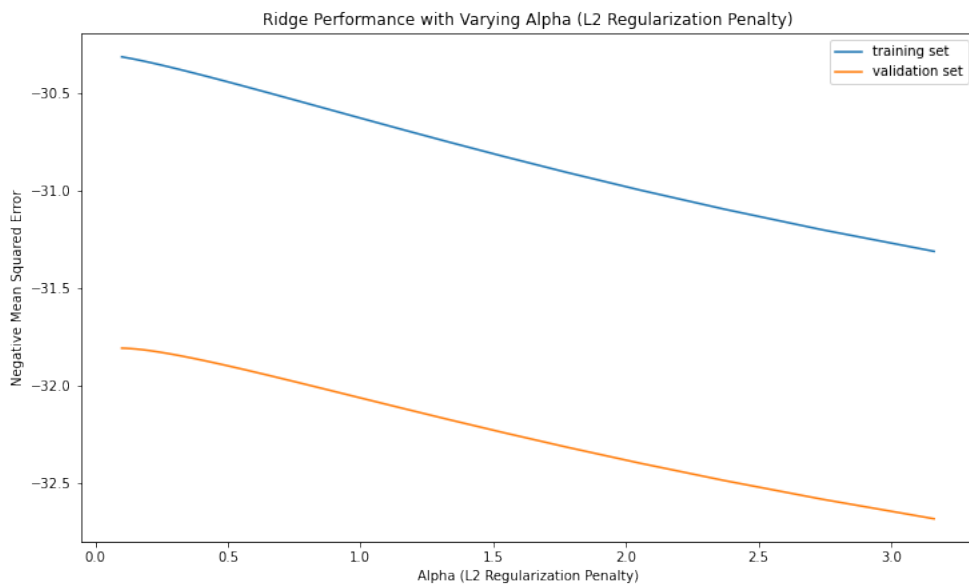


Figure 15. Medal Score Modeling: Lasso Alpha Hyper-Parameter Cross-validation

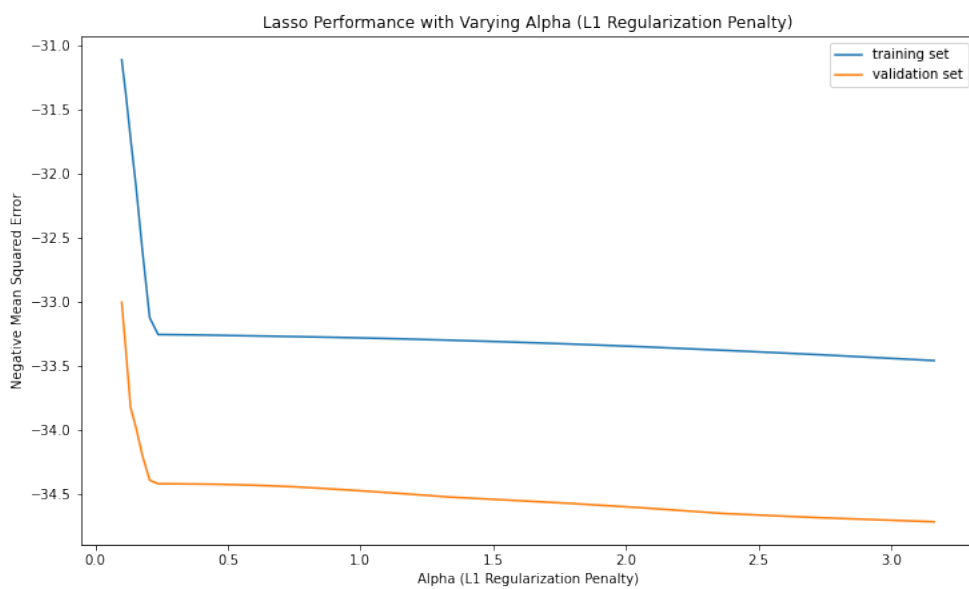


Figure 16. Medal Score Modeling: Regression Trees Hyper-Parameter Cross-validation



Figure 17. Medal Score Modeling: Neural Network Hyper-Parameter Cross-validation

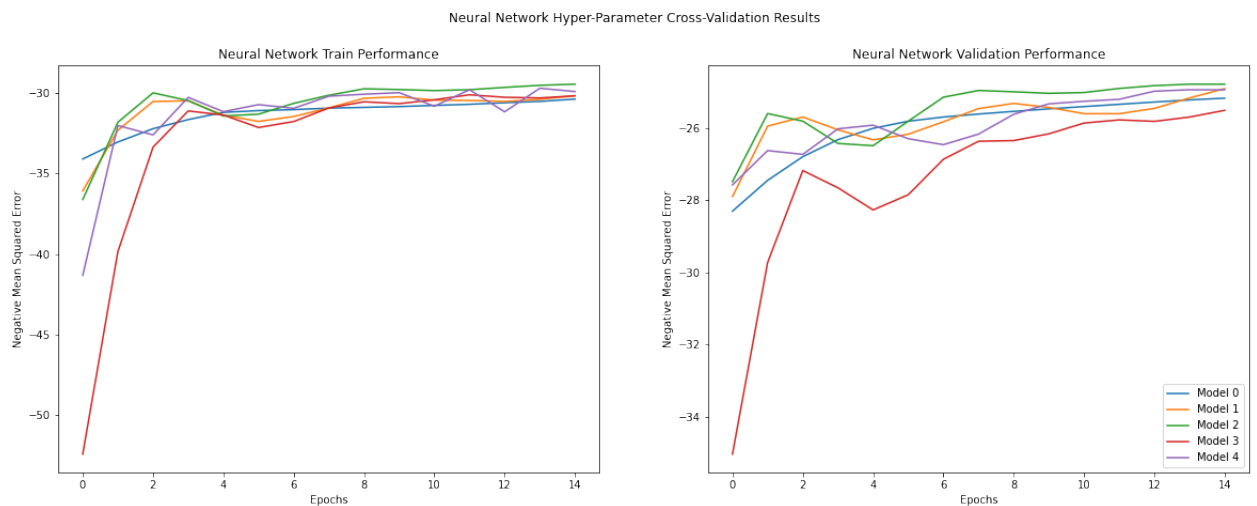


Figure 18. Medal Score Modeling: ANOVA Tables

Medal Score Prediction: Regression Model Summaries and ANOVA Table						
Model Type	Coefficient	Coefficient Value	Standard Error	T-value	P-value	Significance (alpha=0.05)
Linear Regression	Intercept	0.7197	0.361	1.995	0.046813	*
	Pre-Peak Age	0.00188718	0.063	0.03	0.976008	
	Peak Age	0.2116351	0.087	2.428	0.015652	*
	Post-Peak Age	0.14577405	0.039	3.763	0.000195	*
	R-squared	0.2786		F-Statistic	47.7642	
	Adjusted R-Squared	0.2728		p-value	p<0.1e-6	*
Ridge Regression	Intercept	0.7388	0.361	2.046	0.041492	*
	Pre-Peak Age	0.03176011	0.063	0.506	0.613127	
	Peak Age	0.201342	0.087	2.308	0.021547	*
	Post-Peak Age	0.12768363	0.039	3.293	0.001086	*
	R-squared	0.3124		F-Statistic	47.4695	
	Adjusted R-Squared	0.3012		p-value	p<0.1e-6	*
Lasso Regression	Intercept	0.7764	0.361	2.135	0.033438	*
	Pre-Peak Age	0	0.063	0	1	
	Peak Age	0.18800687	0.087	2.157	0.031672	*
	Post-Peak Age	0.15345045	0.039	3.96	0.00009	*
	R-squared	0.2784		F-Statistic	47.7091	
	Adjusted R-Squared	0.2726		p-value	p<0.1e-6	*

Figure 19. Medal Score Modeling: Error Totals

Medal Score Prediction: Model Error Comparison Table				
Model	Partition	Mean Absolute Error	Mean Squared Error	Root Mean Squared Error
Linear Regression	Training	3.1526	30.2833	5.503
	Testing	2.98	25.8711	5.0864
Ridge Regression	Training	3.1739	30.3354	5.5078
	Testing	3.0095	26.0291	5.1019
Lasso Regression	Training	3.1623	30.293	5.5039
	Testing	2.9996	25.9125	5.0904
Regression Trees	Training	3.0688	28.801	5.3667
	Testing	3.061	27.2124	5.2165
Neural Network	Training	2.9273	29.8559	5.4641
	Testing	2.7902	24.9376	4.9938

Figure 20. Individual Dashboard Preview

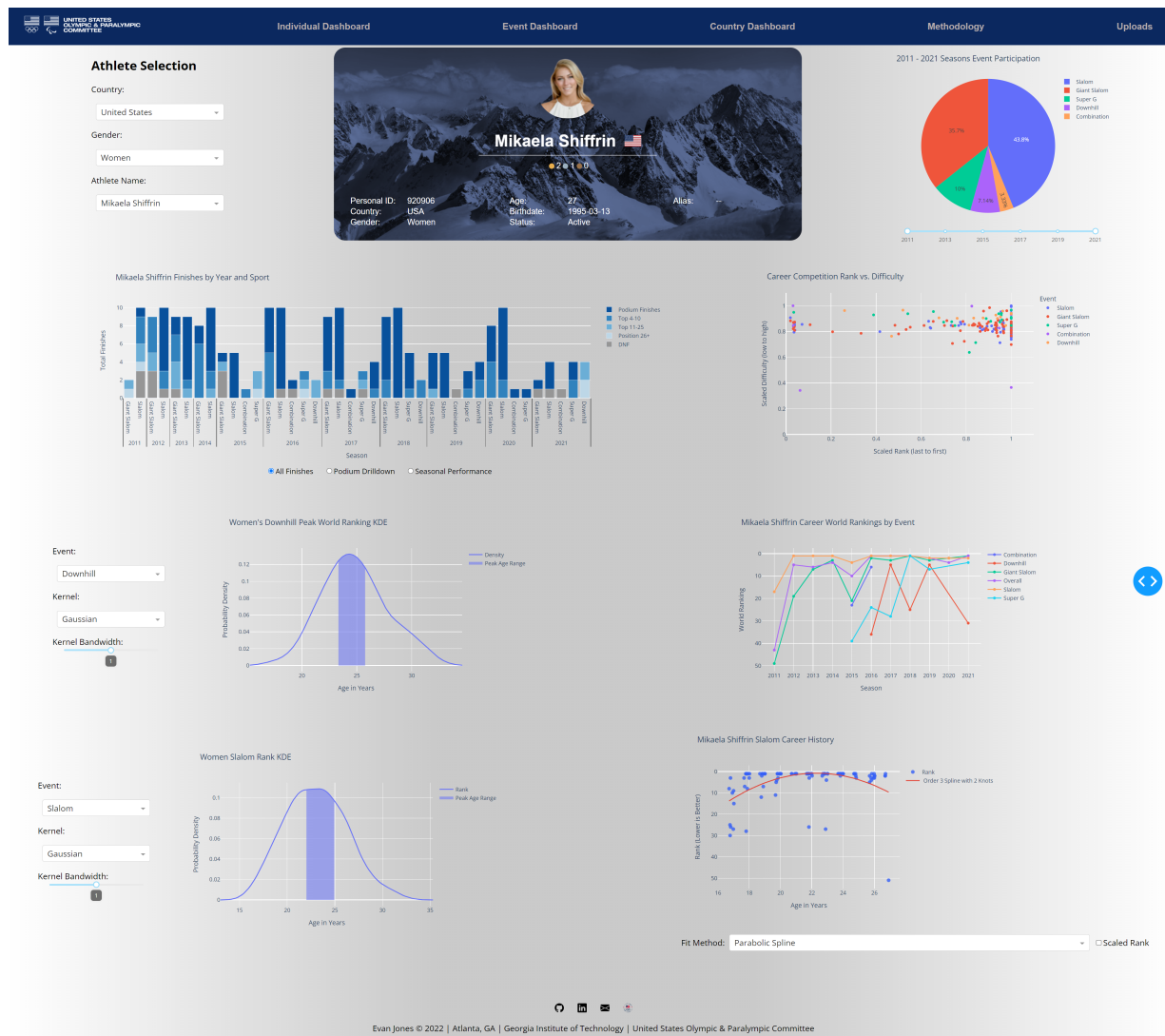


Figure 21. Event Dashboard Preview

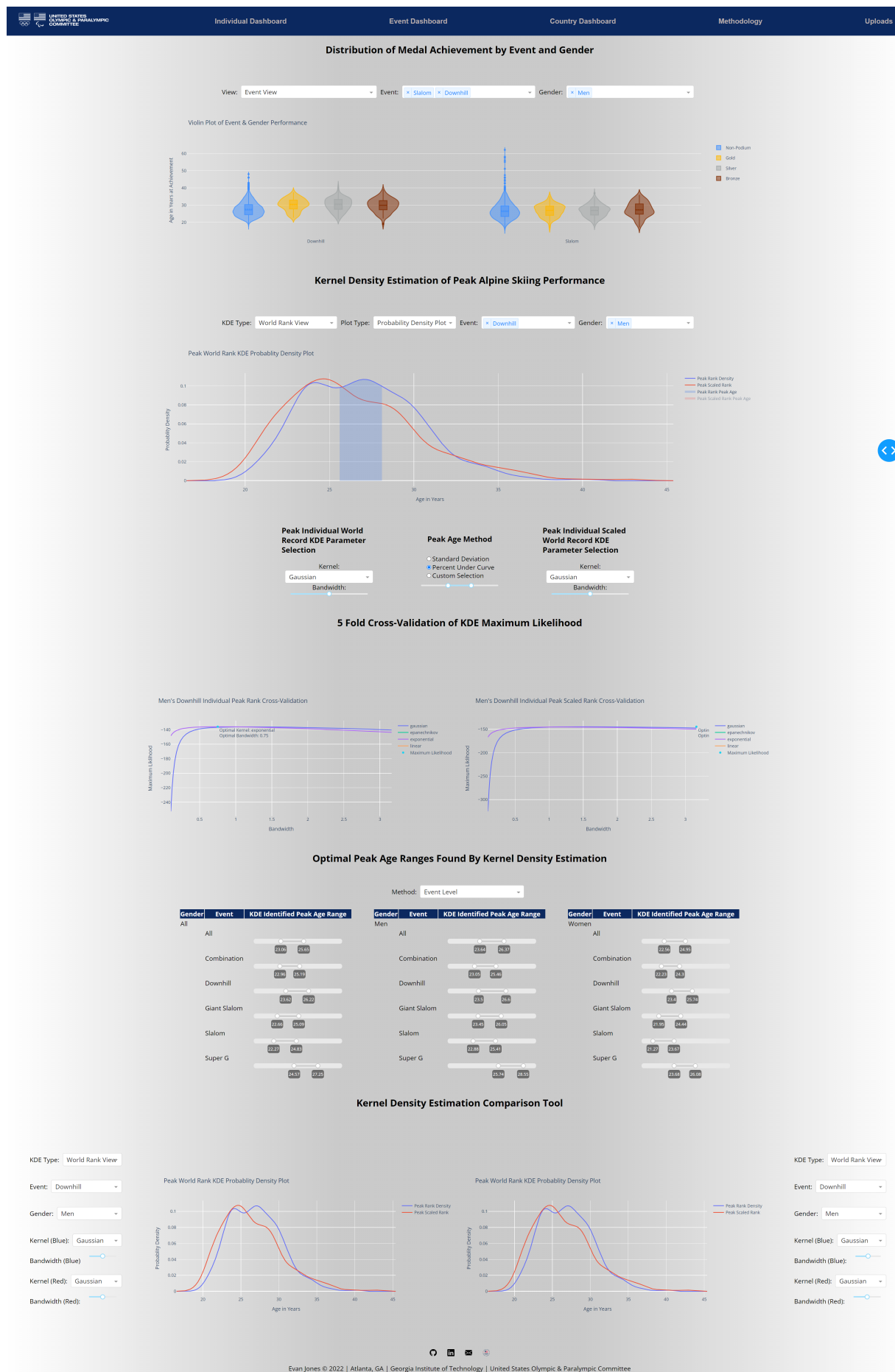


Figure 22. Country/Olympic Dashboard Preview Country Comparison & Pipeline Modeling (1/2)

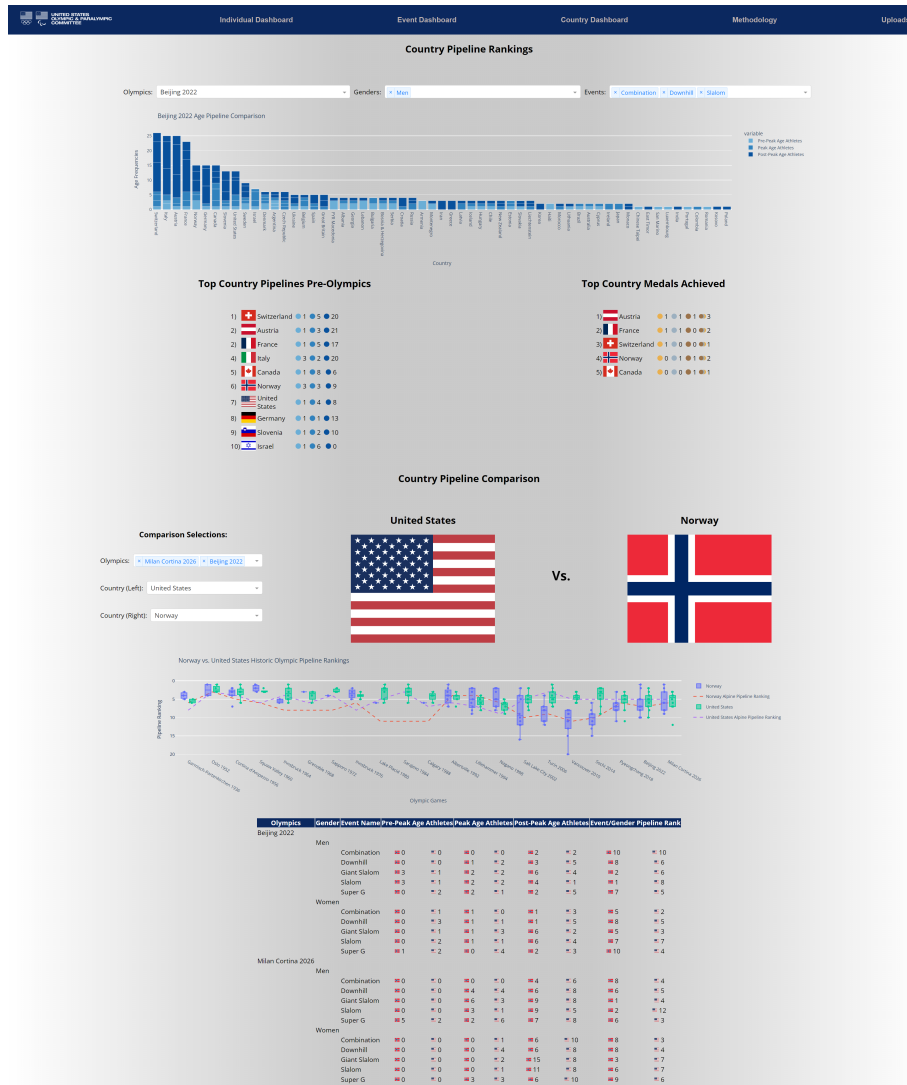


Figure 23. Country/Olympic Dashboard Preview Country Drill-down (2/3)

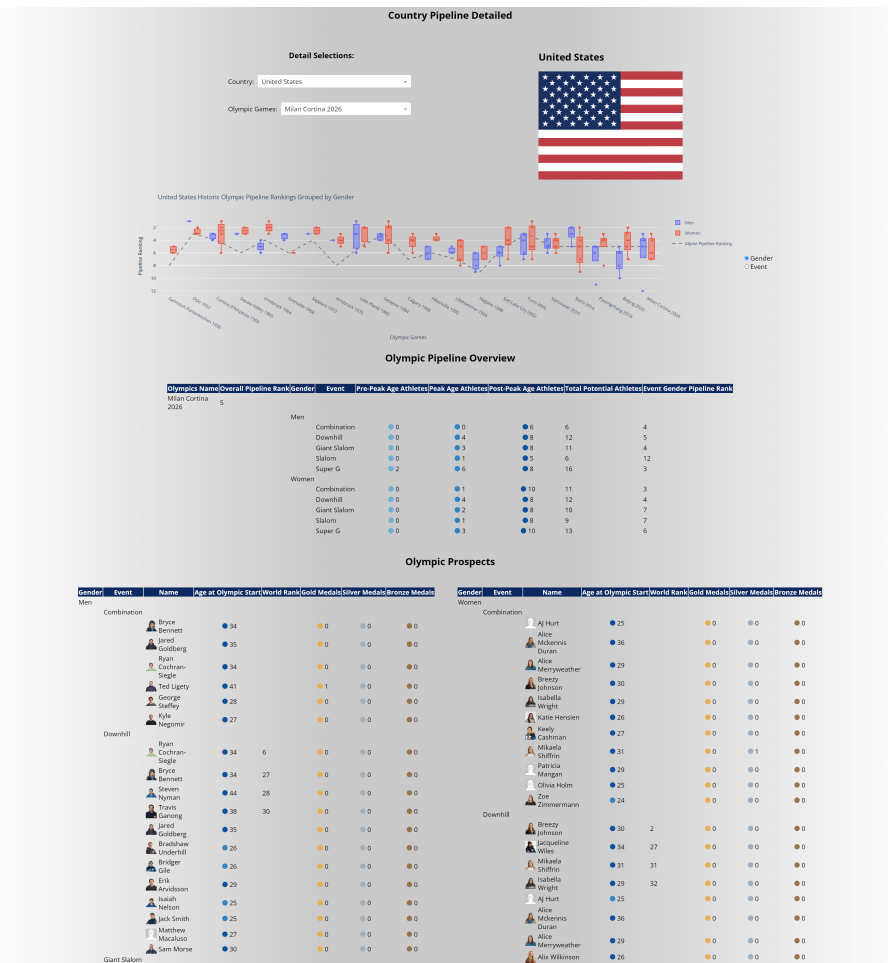


Figure 24. Country/Olympic Dashboard Preview Medal Prediction (3/3)

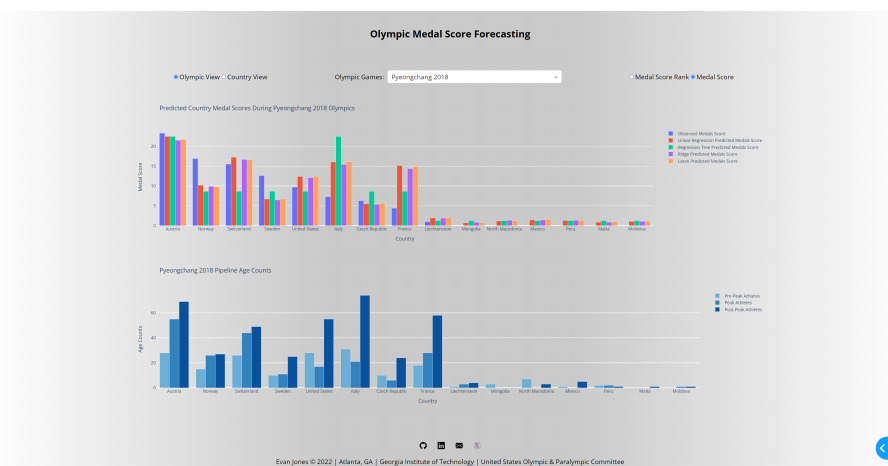


Figure 25. Data Scraper: Generator Diagram

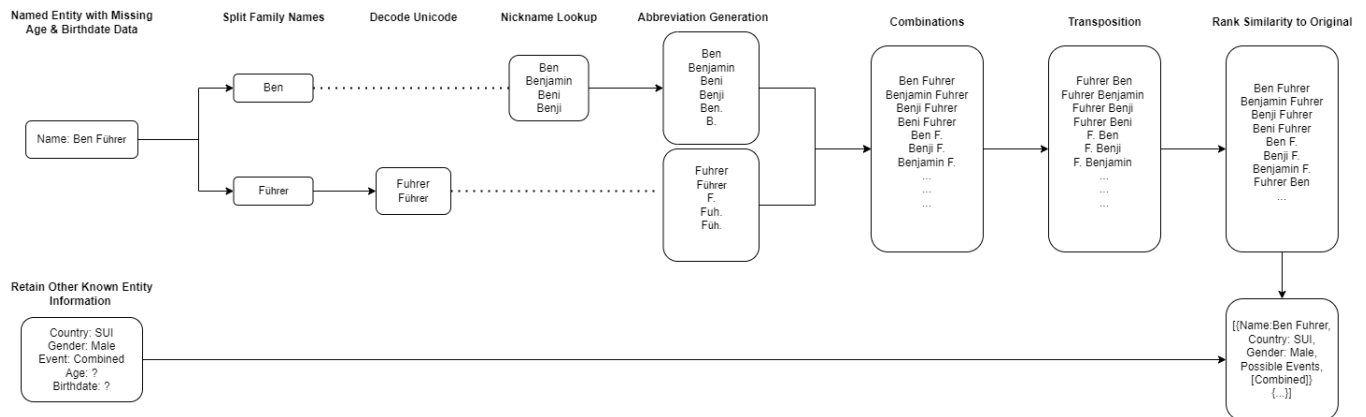


Figure 26. Data Scraper: Discriminator Diagram with Robust Word Embedding

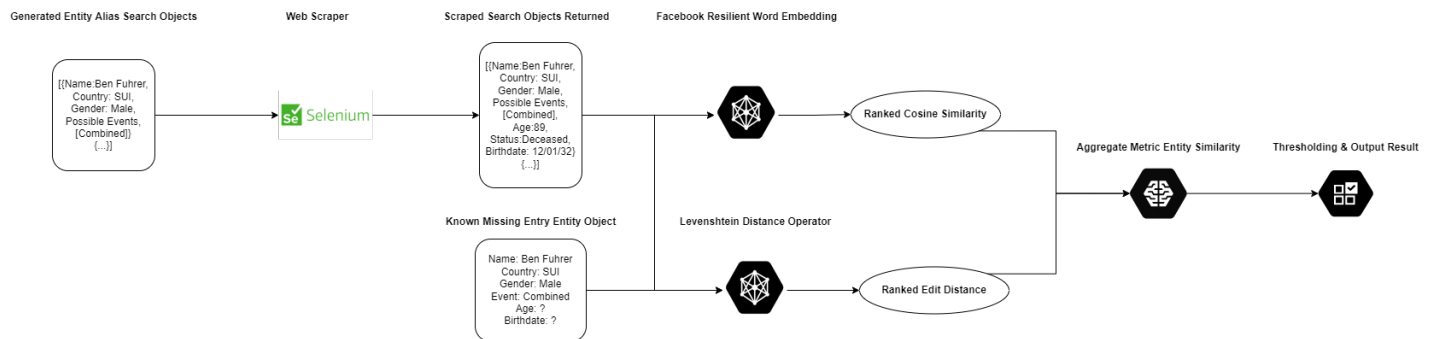


Figure 27. Data Scraper: Discriminator Diagram without Robust Word Embedding

